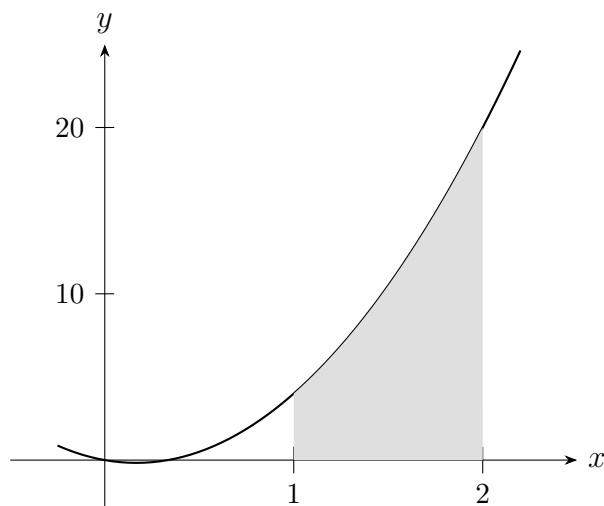


### 5.7 Classwork: Calculus Practice (no calculator)

1. Let  $f(x) = x^3 - 4x + 1$ .
  - (a) Find  $f'(x)$ . [2]
  - (b) Find the gradient of the graph of  $f$  at  $x = 2$ . [2]
  - (c) Find the equation of the tangent line to the graph of  $f$  at  $x = 2$ . Give your answer in the form  $y = mx + c$ . [3]
  - (d) Find the coordinates of the points where the gradient of  $f$  is zero. [2]



- (a) Find  $\int (6x^2 - 2x) dx$ . [3]
- (b) Find the area of the shaded region bounded by the graph of  $f$ , the  $x$ -axis, and the lines  $x = 1$  and  $x = 2$ . [3]
- (c) The graph of  $f$  passes through the point  $(1, 4)$ . A function  $g$  has derivative  $g'(x) = 6x^2 - 2x$  and passes through the point  $(1, 7)$ .  
Find  $g(x)$ . [4]

3. Let  $f(x) = (2x + 1)e^x$ .

- Show that  $f'(x) = (2x + 3)e^x$ . [3]
- Find the equation of the tangent to the graph of  $f$  at the point where  $x = 0$ . [3]
- The tangent found in part (b) intersects the  $x$ -axis at point  $A$ . Find the coordinates of  $A$ . [2]
- Find the  $x$ -coordinate of the point where the graph of  $f$  has a local minimum. Justify that it is a minimum. [4]

4. A particle moves along a straight line. Its velocity  $v \text{ m s}^{-1}$  at time  $t$  seconds is given by

$$v(t) = t^2 - 5t + 4, \quad \text{for } t \geq 0.$$

When  $t = 0$ , the particle is at position  $s = 2$  m from the origin.

- Write down the velocity of the particle when  $t = 0$ . [1]
- Find the times when the particle is at rest. [3]
- Find the acceleration of the particle when  $t = 3$ . [3]
- Find an expression for the displacement  $s(t)$ . [4]
- Find the total distance travelled by the particle in the first 4 seconds. [3]